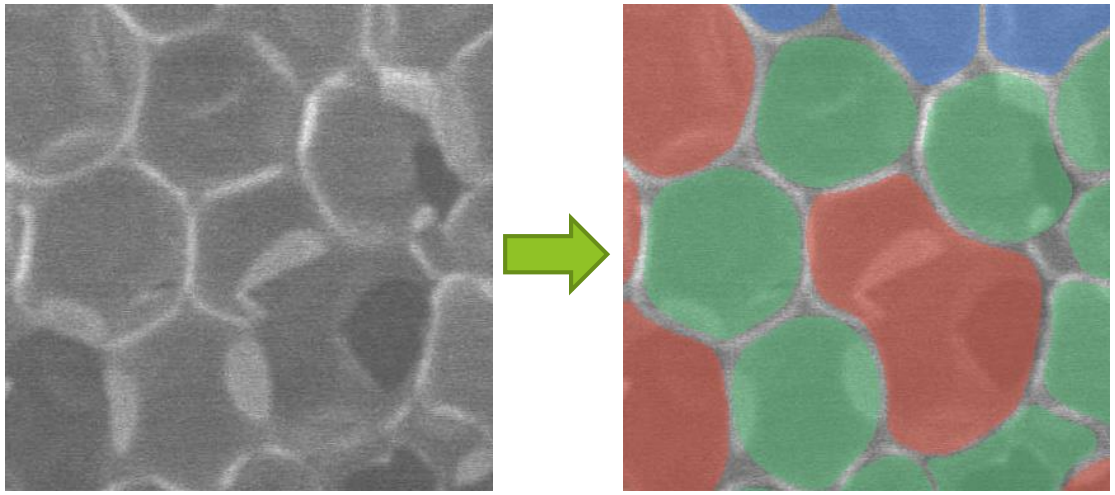
2026 ML Final Competition Intro.

2026/10/29

Task Description

- Task:
  - Image segmentation of foaming material SEM images.
  - Classifies the size of each foams
- Training dataset: 78 SEM images with **4-class labels (0-3, ignore 255)**, and 366 unlabeled SEM images(up to 1536 × 1103).
- Evaluation dataset: **113** SEM images (384 × 384).



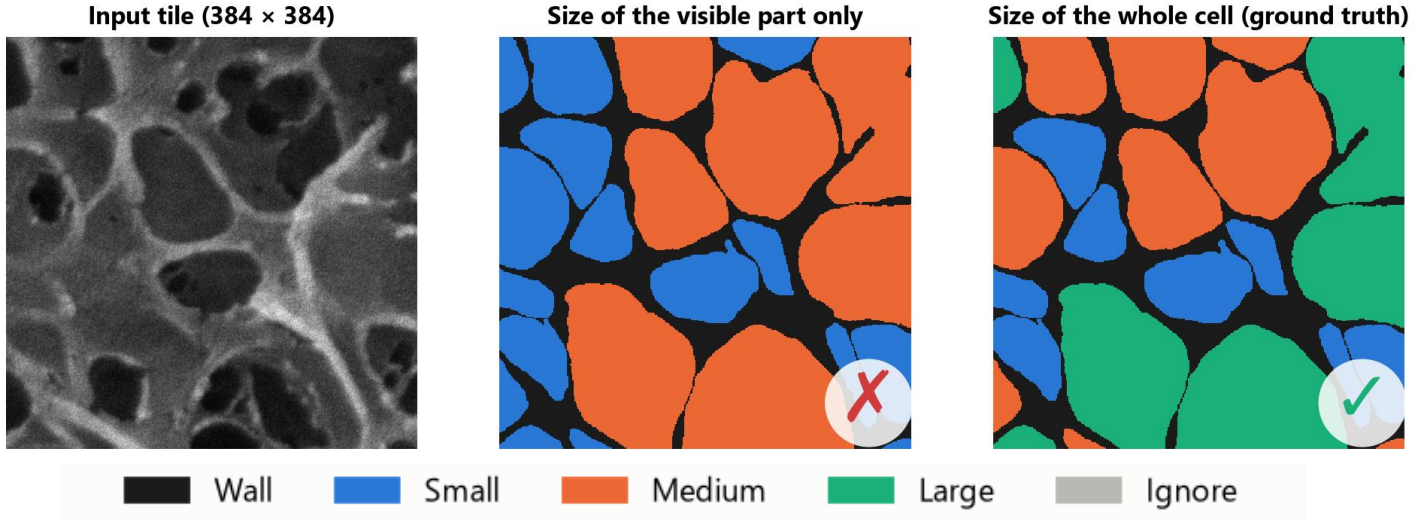
3 (Red) Large

2 (Green) Medium

1 (Blue) Small

0 for otherwise

Task Description

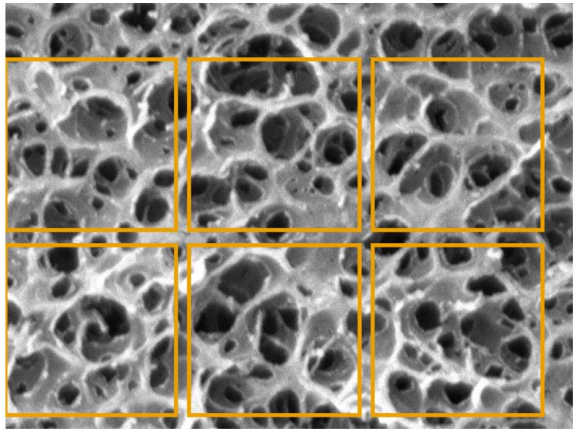


Its size class **is decided by its whole area in the original SEM image**, even when figure shows only part of it.

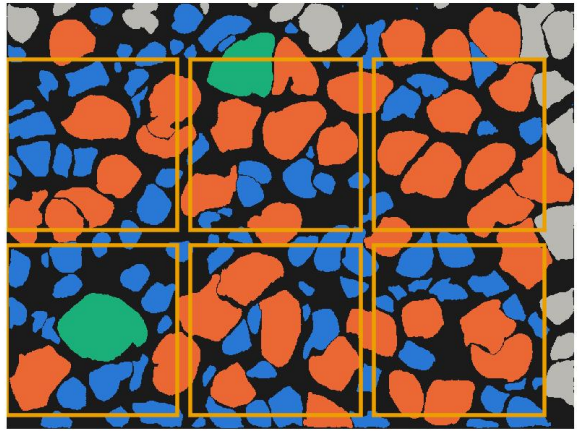
Type	True Size(pixel)	label	% in dataset
Wall	x	0	0.289
Small	< 6000	1	0.180
Medium	6000 ~ 14000	2	0.170
Large	> 14000	3	0.283
Ignore	x	255	0.077

Dataset

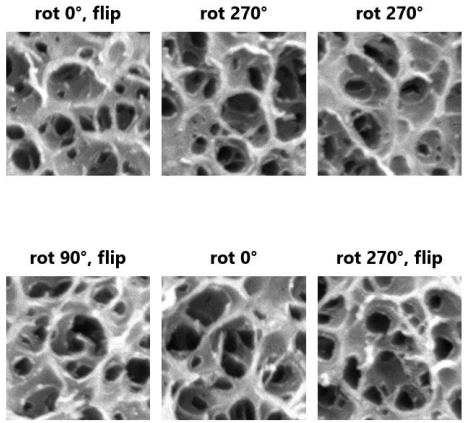
Original SEM image (1280 × 960): 2 × 3 tiles



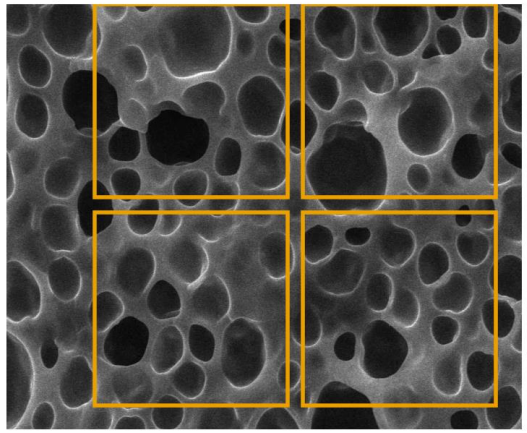
Its label (cell size from the whole image)



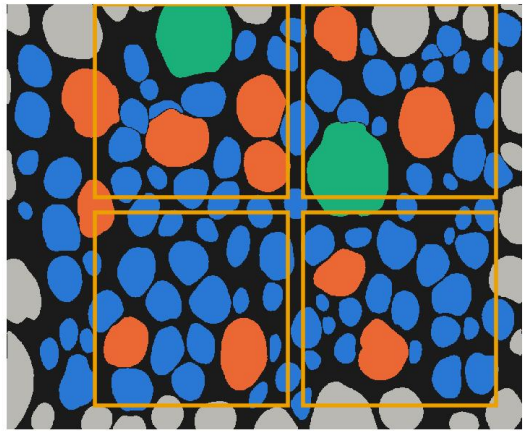
Tiles as given (rotated / flipped)

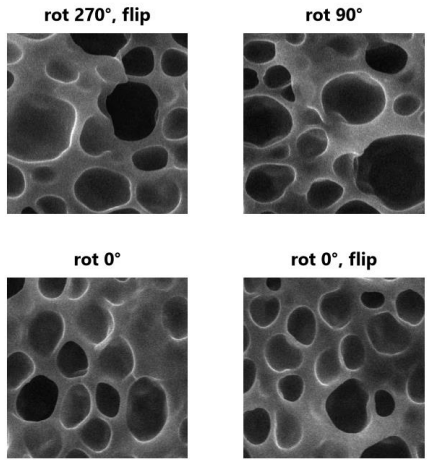


Original SEM image (1024 × 846): 2 × 2 tiles



Its label (cell size from the whole image)





Wall (0) Small (1) Medium (2) Large (3) Ignore (255) Tile (384 × 384)

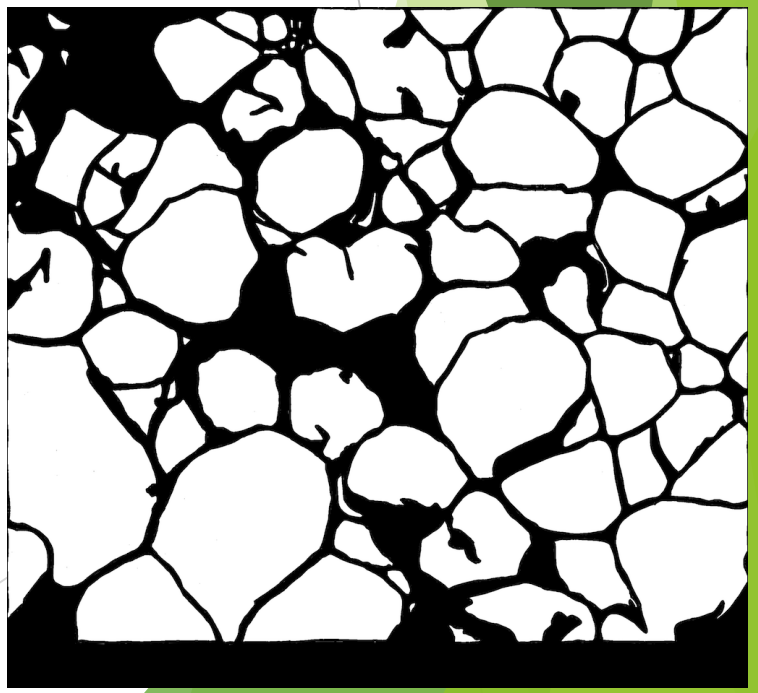
Evaluation

- Input image size: 384 × 384
- Input channel: Grayscale
- Output: 4-class(0/1/2/3) mask with 384 × 384
- mean IoU (intersection over union) over the three cell classes (small, medium, large), averaged over the evaluation tiles.(6 images for public, and the other for private).

$$IoU_c = \frac{|y_c \cap \hat{y}_c|}{|y_c \cup \hat{y}_c|}, score = \frac{1}{C}\sum_{c \subset C} IoU_c$$

- $\hat{y}_c$is the set of pixels of the predicted mask as class c.
- y_c is the set of pixels of the groundtruth mask as class c.





Grading Policy

- 等第+名次取高分